

MATA37H3 — Week 01

Sigma Notation, Riemann Sums & the Definite Integral

Recap and orientation

This unit builds the machinery needed to give a rigorous meaning to “the area under a curve,” which is exactly what the definite integral will turn out to be. We start with **sigma notation**, a compact way to write long sums, then use sums of rectangle areas (**Riemann sums**) to approximate area, and finally take a limit as the number of rectangles goes to infinity to *define* the definite integral. Along the way we introduce **partitions**, **upper and lower sums**, and the supremum/infimum ideas that make the definition precise. Everything in Units 8 onward (the Fundamental Theorem of Calculus, integration techniques) builds directly on the definitions in this unit.

Why this matters

Every integral you will ever compute is, underneath the antiderivative shortcuts you’ll learn in Unit 8, defined as a limit of sums of rectangle areas. Understanding *why* that limit exists and *what it means* for a function to be integrable is what separates knowing the mechanics of integration from actually understanding calculus — and it is a favourite source of “trap” questions on rigorous exams (is a given set a partition? does a given expression equal $L_P(f)$ or $U_P(f)$? does the limit exist here?).

DEFINITION Sigma (Summation) Notation (OpenStax §1.1)

The Greek capital letter Σ is used to write a long sum compactly. In the form

$$\sum_{i=1}^n a_i,$$

a_i describes the terms being added and i is called the **index**. We evaluate the term at $i = 1$, then $i = 2$, and so on up through $i = n$, and add all the resulting values. The index is a **dummy variable**: it exists only to keep track of which terms to add, and does not affect the value of the sum. We may use any letter we like for the index ($i, j, k, n, \dots$), and starting the sum at a different lower limit (e.g. $\sum_{i=2}^4$) simply changes which terms are included.

KEY FACT Properties of Sigma Notation (OpenStax §1.1)

Let $a_1, \dots, a_n$ and $b_1, \dots, b_n$ be two sequences of terms and let c be a constant. For all positive integers n and integers m with $1 \leq m \leq n$:

$$\begin{aligned} \sum_{i=1}^n c &= nc, & \sum_{i=1}^n ca_i &= c \sum_{i=1}^n a_i, \\ \sum_{i=1}^n (a_i + b_i) &= \sum_{i=1}^n a_i + \sum_{i=1}^n b_i, & \sum_{i=1}^n (a_i - b_i) &= \sum_{i=1}^n a_i - \sum_{i=1}^n b_i, \\ \sum_{i=1}^n a_i &= \sum_{i=1}^m a_i + \sum_{i=m+1}^n a_i. \end{aligned}$$

The last property lets you split a sum at any point and is exactly what makes **telescoping sums** collapse.

KEY FACT Sums and Powers of Integers (OpenStax §1.1)

$$\sum_{i=1}^n i = \frac{n(n+1)}{2}, \quad \sum_{i=1}^n i^2 = \frac{n(n+1)(2n+1)}{6}, \quad \sum_{i=1}^n i^3 = \frac{n^2(n+1)^2}{4}.$$

These closed-form formulas are what let you evaluate a Riemann sum *exactly*, in closed form, before taking the limit as $n \rightarrow \infty$ (rather than only estimating it numerically).

DEFINITION Partition (OpenStax §1.1)

A set of points $P = \{x_0, x_1, \dots, x_n\}$ with $a = x_0 < x_1 < x_2 < \dots < x_n = b$, dividing $[a, b]$ into the subintervals $[x_0, x_1], [x_1, x_2], \dots, [x_{n-1}, x_n]$, is called a **partition** of $[a, b]$. If every subinterval has the same width, P is a **regular partition**, and that common width is denoted $\Delta x = \frac{b-a}{n}$. For an arbitrary (possibly irregular) partition, the width of the i -th subinterval is written $\Delta x_i = x_i - x_{i-1}$.

Watch for: a partition of $[a, b]$ must *always* include both endpoints a and b — a finite set of points strictly inside (a, b) is not a partition of $[a, b]$ on its own.

KEY FACT Left- and Right-Endpoint Approximation (OpenStax §1.1)

On each subinterval $[x_{i-1}, x_i]$ of a regular partition, build a rectangle of width Δx . Using the **left** endpoint's function value for the height gives the left-endpoint approximation; using the **right** endpoint's value gives the right-endpoint approximation:

$$L_n = \sum_{i=1}^n f(x_{i-1})\Delta x, \quad R_n = \sum_{i=1}^n f(x_i)\Delta x.$$

Both are estimates of the area under $y = f(x)$ on $[a, b]$; neither is exact for finite n , but both improve as n grows.

DEFINITION Riemann Sum (OpenStax §1.1)

Let f be defined on $[a, b]$ and let P be a partition with subintervals of width Δx_i . For each i , choose **any** point x_i^* in $[x_{i-1}, x_i]$ (a “sample point” — not necessarily an endpoint). The corresponding **Riemann sum** for f is

$$\sum_{i=1}^n f(x_i^*) \Delta x_i.$$

Left- and right-endpoint approximations are the special cases $x_i^* = x_{i-1}$ and $x_i^* = x_i$.

DEFINITION Area via a Limit of Riemann Sums (OpenStax §1.1)

If f is continuous and nonnegative on $[a, b]$, the (exact) area under $y = f(x)$ on $[a, b]$ is

$$A = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x.$$

This limit can be shown to exist and to be the same regardless of how the sample points x_i^* are chosen, whenever f is continuous — this independence-of-choice is what makes “the area under the curve” a well-defined number rather than something that depends on an arbitrary choice.

KEY FACT Upper Sums, Lower Sums, and $L_P(f)$, $U_P(f)$ (OpenStax §1.1)

If, on each subinterval, x_i^* is chosen so that $f(x_i^*)$ is the **largest** value of f on $[x_{i-1}, x_i]$, the resulting Riemann sum is called the **upper sum**, written $U_P(f)$. If $f(x_i^*)$ is instead the **smallest** value of f on that subinterval, the resulting sum is the **lower sum**, $L_P(f)$. (Since

f is continuous, it does attain a largest and smallest value on each closed subinterval.) For any partition P and any such f ,

$$L_P(f) \leq U_P(f),$$

because every lower estimate on a subinterval is $\leq$ every upper estimate on that same subinterval. If f is monotonic (increasing or decreasing throughout $[a, b]$), the maximum/minimum on each subinterval automatically occurs at an endpoint, so $L_P(f)$ and $U_P(f)$ coincide with whichever of L_n, R_n is the underestimate/overestimate.

Refining a partition: if $Q \supseteq P$ (i.e. Q is obtained from P by adding more partition points — a **finer** partition), then

$$L_P(f) \leq L_Q(f) \leq U_Q(f) \leq U_P(f).$$

Lower sums only ever increase and upper sums only ever decrease as you add more points — they “squeeze” toward each other.

DEFINITION The Definite Integral (OpenStax §1.2)

If f is a function defined on $[a, b]$, the **definite integral** of f from a to b is

$$\int_a^b f(x) dx = \lim_{n \rightarrow \infty} \sum_{i=1}^n f(x_i^*) \Delta x,$$

provided this limit exists (and is the same for every choice of sample points x_i^*). If it exists, f is called **integrable** on $[a, b]$. Here a, b are the **limits of integration** (a lower, b upper), $f(x)$ is the **integrand**, and x (in dx) is the **variable of integration** — itself a dummy variable, so $\int_a^b f(x) dx = \int_a^b f(t) dt$.

KEY FACT Continuous Functions Are Integrable (OpenStax §1.2, Theorem 1.1)

If f is continuous on $[a, b]$, then f is integrable on $[a, b]$. (Integrability can still hold for some discontinuous functions too — e.g. a function with finitely many jump discontinuities — but continuity is the simplest sufficient condition, and the one used most often in this course.)

Worked Examples

EXAMPLE 1

Evaluate a sum using sigma-notation properties

Evaluate $\sum_{i=1}^{50} (i^2 - 4i + 7)$.

Split the sum using the sigma-notation properties, then apply the closed-form formulas:

$$\sum_{i=1}^{50} (i^2 - 4i + 7) = \sum_{i=1}^{50} i^2 - 4 \sum_{i=1}^{50} i + \sum_{i=1}^{50} 7.$$

Using $\sum i^2 = \frac{n(n+1)(2n+1)}{6}$ and $\sum i = \frac{n(n+1)}{2}$ with $n = 50$:

$$\sum_{i=1}^{50} i^2 = \frac{50(51)(101)}{6} = 42,925, \quad \sum_{i=1}^{50} i = \frac{50(51)}{2} = 1275, \quad \sum_{i=1}^{50} 7 = 50(7) = 350.$$

So the sum equals $42,925 - 4(1275) + 350 = 42,925 - 5100 + 350 = 38,175$. ■

EXAMPLE 2

Telescoping sum

Evaluate $\sum_{i=1}^{99} \left(\frac{1}{i} - \frac{1}{i+1} \right)$ exactly.

Writing out the first few and last few terms,

$$\left(\frac{1}{1} - \frac{1}{2} \right) + \left(\frac{1}{2} - \frac{1}{3} \right) + \left(\frac{1}{3} - \frac{1}{4} \right) + \cdots + \left(\frac{1}{98} - \frac{1}{99} \right) + \left(\frac{1}{99} - \frac{1}{100} \right).$$

Every term of the form $-\frac{1}{k}$ is exactly cancelled by the $+\frac{1}{k}$ in the next bracket, except the

very first $+\frac{1}{1}$ and the very last $-\frac{1}{100}$. So the whole sum **collapses** to

$$1 - \frac{1}{100} = \frac{99}{100}. \quad \blacksquare$$

This is the “telescoping” trick: whenever a sum’s general term is a *difference* of consecutive values of some expression, splitting it via the sigma-notation properties and writing out a few terms reveals the cancellation pattern.

EXAMPLE 3

Left- and right-endpoint approximation

Approximate the area under $f(x) = x^2 + 1$ on $[0, 2]$ using $n = 4$ subintervals, with both the left- and right-endpoint approximations.

With $n = 4$, $\Delta x = \frac{2-0}{4} = 0.5$, so the partition points are $x_0 = 0$, $x_1 = 0.5$, $x_2 = 1$, $x_3 = 1.5$, $x_4 = 2$, and $f(x_0) = 1$, $f(x_1) = 1.25$, $f(x_2) = 2$, $f(x_3) = 3.25$, $f(x_4) = 5$.

Left endpoints x_0, x_1, x_2, x_3 :

$$L_4 = (1 + 1.25 + 2 + 3.25)(0.5) = (7.5)(0.5) = 3.75.$$

Right endpoints x_1, x_2, x_3, x_4 :

$$R_4 = (1.25 + 2 + 3.25 + 5)(0.5) = (11.5)(0.5) = 5.75.$$

Since f is increasing on $[0, 2]$, L_4 underestimates and R_4 overestimates the true area, and $L_4 = L_P(f) \leq U_P(f) = R_4$ for this partition, as the theory predicts. $\blacksquare$

EXAMPLE 4

Upper and lower sums on a given partition

Let $f(x) = 9 - x^2$ on $[0, 2]$, and let $P = \{0, 1, 2\}$. Compute $L_P(f)$ and $U_P(f)$.

f is **decreasing** on $[0, 2]$ (since $f'(x) = -2x < 0$ for $x > 0$), so on each subinterval the maximum of f occurs at the *left* endpoint and the minimum at the *right* endpoint.

Subinterval $[0, 1]$: $f(0) = 9$ (max), $f(1) = 8$ (min), width $\Delta x_1 = 1$. Subinterval $[1, 2]$: $f(1) = 8$ (max), $f(2) = 5$ (min), width $\Delta x_2 = 1$.

$U_P(f) = f(0)(1) + f(1)(1) = 9 + 8 = 17$, $L_P(f) = f(1)(1) + f(2)(1) = 8 + 5 = 13$.
Indeed $L_P(f) = 13 \leq 17 = U_P(f)$. $\blacksquare$

EXAMPLE 5

Refining a partition

Using f from Example 4, refine $P = \{0, 1, 2\}$ to $Q = \{0, 0.5, 1, 2\}$ and recompute $L_Q(f)$, $U_Q(f)$. Confirm $L_P(f) \leq L_Q(f) \leq U_Q(f) \leq U_P(f)$.

f is still decreasing, so on each subinterval of Q the left endpoint gives the max and the right endpoint the min.

Subintervals: $[0, 0.5]$ ($\Delta x = 0.5$): $f(0) = 9$, $f(0.5) = 8.75$. $[0.5, 1]$ ($\Delta x = 0.5$): $f(0.5) = 8.75$, $f(1) = 8$. $[1, 2]$ ($\Delta x = 1$): $f(1) = 8$, $f(2) = 5$.

$$U_Q(f) = 9(0.5) + 8.75(0.5) + 8(1) = 4.5 + 4.375 + 8 = 16.875,$$

$$L_Q(f) = 8.75(0.5) + 8(0.5) + 5(1) = 4.375 + 4 + 5 = 13.375.$$

Comparing to Example 4 ($L_P = 13$, $U_P = 17$): $13 \leq 13.375 \leq 16.875 \leq 17$. ✓ The lower sum increased and the upper sum decreased when we refined the partition, exactly as the Key Fact predicts. ■

EXAMPLE 6

Evaluate a definite integral from the limit definition

Use the definition of the definite integral (with right-endpoint sample points) to evaluate

$$\int_0^1 x^2 dx.$$

Here $a = 0$, $b = 1$, so $\Delta x = \frac{1-0}{n} = \frac{1}{n}$, and the right endpoint of the i -th subinterval is

$x_i = 0 + i\Delta x = \frac{i}{n}$. Then

$$f(x_i) = x_i^2 = \frac{i^2}{n^2}, \quad \sum_{i=1}^n f(x_i)\Delta x = \sum_{i=1}^n \frac{i^2}{n^2} \cdot \frac{1}{n} = \frac{1}{n^3} \sum_{i=1}^n i^2 = \frac{1}{n^3} \cdot \frac{n(n+1)(2n+1)}{6}.$$

Expanding and simplifying,

$$\frac{n(n+1)(2n+1)}{6n^3} = \frac{2n^3+3n^2+n}{6n^3} = \frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2}.$$

Taking the limit as $n \rightarrow \infty$, the last two terms vanish:

$$\int_0^1 x^2 dx = \lim_{n \rightarrow \infty} \left(\frac{1}{3} + \frac{1}{2n} + \frac{1}{6n^2} \right) = \frac{1}{3}. \quad \blacksquare$$

EXAMPLE 7

Deciding whether a set is a partition

Which of the following are valid partitions of $[0, 3]$: (a) $\{0, 3\}$; (b) $\{0, 1, 1, 2, 3\}$; (c) $\{1, 3\}$; (d) $\{0, \sqrt{2}, 3\}$?

A partition of $[0, 3]$ must be a set of points that includes **both endpoints**, listed in strictly increasing order (repeats are meaningless in a set, and the points don't need to be evenly spaced). (a) $\{0, 3\}$ — valid; it's the trivial partition with a single subinterval $[0, 3]$. (b) As a set, $\{0, 1, 1, 2, 3\} = \{0, 1, 2, 3\}$ — valid (the repeated 1 doesn't create a second point). (c) $\{1, 3\}$ — **not** valid: it does not include the left endpoint 0, so it isn't a partition of $[0, 3]$ (it would be a valid partition of $[1, 3]$). (d) $\{0, \sqrt{2}, 3\}$ — valid; a partition's points don't need to be rational or evenly spaced, only strictly increasing and equal to a, b at the ends. ■

Practice Problems

Read the hint and the caution before you start each problem — not after.

Sigma Notation & Sums

1. Evaluate $\sum_{i=1}^6 (2i-3)$ two ways: (a) by expanding term by term, and (b) by first splitting the sum using the sigma-notation properties. Confirm the two methods agree.

HINT Split the sum using the sigma-notation properties from Page 1, then plug $n = 6$ into the closed-form formulas for $\sum i$ and $\sum 1$.

CAUTION — before you solve Don't forget the constant term -3 must be summed n times too — it becomes $-3n$, not just -3 .

Your working:

2. Write $1^3 + 2^3 + 3^3 + \cdots + 40^3$ in sigma notation, then evaluate it using the closed-form formula for $\sum i^3$.

HINT Write the sum in sigma notation first, $\sum_{i=1}^{40} i^3$, then apply the closed-form formula directly with $n = 40$.

CAUTION — before you solve Don't reach for the $\sum i^2$ formula by mistake — make sure you're using $\sum i^3 = \frac{n^2(n+1)^2}{4}$.

Your working:

3. Evaluate $\sum_{i=1}^{250} \left(\frac{1}{i+1} - \frac{1}{i+2} \right)$ exactly, using the telescoping-sum idea.

HINT Write out the first few and last few terms explicitly to see exactly which boundary terms survive the cancellation.

CAUTION — before you solve Watch the indexing carefully — the sum starts at $\frac{1}{i+1}$ with $i = 1$, so the surviving terms are $\frac{1}{2}$ and $-\frac{1}{252}$, not 1 and $-\frac{1}{251}$.

Your working:

4. Compute $\sum_{i=1}^N \sum_{k=1}^i 1$ and $\sum_{i=1}^N \sum_{k=1}^i k$ in terms of N . (Evaluate the inner sum first for each fixed i , then sum the result over i .)

HINT Evaluate the inner sum first for a fixed value of i , then sum the resulting expression over i .

CAUTION — before you solve Don't treat the inner sum's upper limit i as if it were the constant N — it changes with each outer term.

Your working:

Partitions & Riemann Sums

5. Which of the following are partitions of $[-1, 4]$? (a) $[-1, 4]$ itself (the whole interval, not a finite set of points); (b) $\{-1, 0, 2, 4\}$; (c) $\{-1, 4\}$; (d) $\{0, 2, 4\}$; (e) $\left\{-1 + \frac{5n}{n+1} : n \in \mathbb{N}\right\} \cup \{4\}$.

HINT A partition of $[-1, 4]$ must be a finite set of points including both endpoints, listed in strictly increasing order.

CAUTION — before you solve Check finiteness as well as the endpoints — an infinite set of points is not a partition even if it appears to approach both endpoints.

Your working:

6. Approximate the area under $f(x) = \sqrt{x} + 1$ on $[0, 4]$ using $n = 4$ subintervals, with both the left- and right-endpoint approximations. Which one is the overestimate here, and why?

HINT Compute Δx and the endpoint function values first, then decide which sum is bigger based on whether f is increasing or decreasing.

CAUTION — before you solve Don't assume the right-endpoint sum is automatically the overestimate — that only holds when f is increasing on the interval.

Your working:

7. Let $f(x) = 8 - 2x$ on $[0, 3]$ and let $P = \{0, 1, 3\}$ (an *irregular* partition — the two subintervals do not have equal width). Compute $L_P(f)$ and $U_P(f)$, being careful to use the correct Δx_i for each subinterval.

HINT Compute $\Delta x_i = x_i - x_{i-1}$ separately for each subinterval before multiplying by the max/min function value.

CAUTION — before you solve This partition is irregular — don't use the same Δx for both subintervals.

Your working:

8. Let f be an **increasing**, bounded function on $[a, b]$, and let $P = \{x_0, x_1, \dots, x_N\}$ be a partition. Write the correct summation formula for $L_P(f)$ and for $U_P(f)$ in terms of $f(x_{i-1})$, $f(x_i)$, and Δx_i — and explain why the formulas are the *opposite* way around from the decreasing-function case in the Key Fact box on Page 1.

HINT Think about where the maximum and minimum of f occur on each subinterval when f is increasing, not decreasing.

CAUTION — before you solve Don't just copy the decreasing-function formulas from Page 1 — the roles of left and right endpoints swap for an increasing function.

Your working:

The Definite Integral

9. Use the limit definition of the definite integral (with right-endpoint sample points) to evaluate $\int_0^2 x^2 dx$.

HINT Set up $\Delta x = \frac{2-0}{n}$ and $x_i = \frac{2i}{n}$, then redo the same algebra as the Page 2-3 worked example with the new limits.

CAUTION — before you solve Don't reuse the constant $\frac{1}{3}$ from the $\int_0^1 x^2 dx$ example — the new limits of integration change that constant.

Your working:

10. Suppose $L_P(f) = 4$, $U_P(f) = 10$ for a partition P , and $L_Q(f) = 6$, $U_Q(f) = 9$ for a partition Q . (a) Is it possible that $P \subseteq Q$? Justify using the Key Fact about refining partitions. (b) What can you conclude about the relationship between $L_{P \cup Q}(f)$ and both pairs of values above?

HINT Compare the given numbers against the refining-partition inequality $L_P \leq L_Q \leq U_Q \leq U_P$ to see whether they're consistent with $P \subseteq Q$.

CAUTION — before you solve Being consistent with a containment relationship doesn't prove it — don't conclude $P \subseteq Q$ is forced just because the inequalities happen to hold.

Your working:
